Document Control

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1. Purpose

1.1. The purpose of this document is to detail the process for ventilation of neonates

1.2. The policy applies to all staff involved in the care of the neonate requiring ventilation at NDDH.

1.3. Implementation of this policy will ensure that:

- We can support the safe care of a neonate requiring ventilation
- The risks associated with the management of a neonate requiring ventilation are reduced

2. Definitions

2.1. IPPV - Intermittent positive pressure ventilation

A set pressure is delivered a set number of times a minute irrespective of the baby's own breathing. Rarely used unless an infant is paralysed or lacks respiratory drive.

2.2. CMV - Continuous Mechanical Ventilation

A set pressure is delivered a set number of times a minute irrespective of the baby's own breathing. Rarely used unless an infant is paralysed or lacks respiratory drive.

2.3. CPAP - Continuous Positive Airways Pressure

Is the use of continuous positive pressure to maintain a continuous level of positive airway pressure in a spontaneously breathing baby. This is our first line mode of support for infants with respiratory distress from any cause. The use of early CPAP is effective in reducing the need for ventilation and surfactant in preterm infants with respiratory distress syndrome, as well as being an effective mode of ventilation for transient tachypnoea of the newborn and meconium aspiration syndrome in term and near term infants. An added advantage is that it is also an effective treatment for apnoea.

2.4. I:E - Inspiration: expiration

2.5. MAP - Mean airway pressure

2.6. MV - Minute volume = Tidal volume x Respiratory rate

Minute volume is the amount of gas inhaled or exhaled from a person's lungs in one minute.

2.7. PEEP – Positive end expiratory pressure.

Is the pressure in the lungs above atmospheric pressure that exists at the end of expiration.

2.8. PIP - Peek inspiratory pressure

2.9. Pmax

The safe maximum pressure (set manually) that the ventilator will not exceed to achieve calculated tidal volume. (Usually 5 above the average PIP delivered to achieve the tidal volume). If the ventilator cannot deliver the VT at the Pmax pressure prescribed, it will alarm "Low Tidal Volume". The 'High Pressure Alarm' value will default to 5 above the Pmax setting.

2.10. PSV - Pressure Support Ventilation

The inspiratory time is limited according to the infant's own lung inflation. Therefore the infant is controlling the duration of breaths according to lung mechanics.

2.11. SIMV - Synchronised Intermittent Mandatory Ventilation

Breaths are delivered synchronised with the infant's breathing. Only the set number of breaths can be delivered, thus if the infant is breathing faster than the set rate those additional breaths are not supported. If the infant breathes less than the set rate, the ventilator will synchronise all the breaths, and deliver some un-triggered breaths. All breaths are supported by PEEP. This mode has advantages for weaning, but those infants who breathe in between supported breaths could become tired.

2.12. PC AC - Pressure control - Assist Control (previously known as SIPPV)

Modes of ventilation synchronised with the infant's breathing. Every patient breath that exceeds a threshold trigger will result in a synchronised mechanical breath being delivered. The infant will receive a set minimum number of breaths per minute. Should the infant breathe above this rate, each additional breathe will also be supported. Weaning in this mode is achieved by reducing the peak pressure or the sensitivity of the trigger. There is a risk of air trapping and air leak due to short expiratory time if the expiratory time is too long, coupled with fast breathing rate.

2.13. VG - Volume Guarantee

The goal of VG is to reduce volutrauma and barotrauma to the baby's lungs. It is thought that by delivering the lowest possible pressure required to deliver the volume will result in less lung injury to the baby.

In this mode, the ventilator is asked to deliver a certain expired tidal volume (VT) – usually 4-6 ml/kg breath, however the ventilator default is 4mls/kg at NDDH SCU and is calculated automatically according to the weight of the baby (entered by user). The ventilator will deliver the calculated volume using with the lowest possible PIP pressure. This is done on a breath-to-breath basis, looking at how much volume was delivered with the previous breath and adjusting the PIP up or down to increase or decrease the VT accordingly.

A maximum PIP and inspiratory time can be set. – If the ventilator cannot deliver the appropriate VT, the ventilator will alarm. This mode does not work if there is a big leak (e.g. over 30 %) as the expired VT will be under-reading.

2.14. SCU - Special Care Unit

2.15. Trigger volume

The volume of gas during inspiration required to trigger a ventilation breath

2.16. Trigger window

The length of time in which a baby can initiate a breath triggering ventilation

2.17. VT - tidal volume.

The amount of gas breathed in or out in one breath (safe values approx 4-6 mls/kg, see above for definition of VG mode)

3. Responsibilities

Role of Paediatric Medical Team

3.1. The Paediatric Medical Team are responsible for:

- Acting in accordance with this guideline
- Ensuring that risks associated with ventilator use are minimised
- Keeping the Tertiary unit updated on condition of infant and delivering collaborative plan of care.

Role of Neonatal Staff Nurse

3.2. The Neonatal Staff Nurse is responsible for:

- Keeping ventilator checked daily and in readiness for use
- Ensuring that settings and the maintenance of the ventilator are in accordance with medical team instructions
- Keeping medical team aware of infants condition
- Observing the baby for associated risks and taking appropriate action
- On-going monitoring and care of the baby supported by a ventilator

4. Background

There are limited randomised trial data which establish that any one ventilation strategy or mode is superior. Many ventilation practices are based on individual experience, types of ventilator used and on evaluating changes in blood gas parameters and the clinical condition of the baby in response to changes in ventilation settings.

This guideline is not intended to be adhered to rigidly, but merely to provide some rationale behind interpretation of blood gases and possible changes to ventilation settings in response.

There are two goals of ventilation:

1. Appropriate oxygenation
2. Appropriate ventilation

Oxygenation is affected by several factors such as the inspired oxygen concentration (FiO2), mean airway pressure (MAP), the area of and diffusion across the gas exchange surface.

Ventilation refers primarily to the amount of carbon dioxide exchanging at the alveolar level. Factors which influence this include the gas exchange surface area and diffusion and the amount of gas able to be moved in and out of the lungs.

4.1. Barnstaple Special Care Unit (SCU) is designated a Level 1 unit and part of the South West Neonatal Network.

It is established to care for:

>32 week gestation twins and Singletons

4.2. It has 2 Intensive Care/High Dependency cots for short term stabilization purposes of the intensive care infant only. Retrieval to Level 2 and 3 neonatal units occurs by use of the Peninsula Neonatal Transport Service (PNTS). This ensures that babies are able to access the right level of care in the right place, proportional to the severity of the illness.

4.3. Due to the rural location of SCU and the distance from PNTS, intensive care babies may need to be stabilized and then managed locally for several hours until the retrieval team arrives.

4.4. Ventilated infants require intensive care management.

4.5. This guideline relates to all staff groups who may be involved in the care of babies requiring ventilation as respiratory support. This includes the Paediatric Medical Team and Registered Neonatal Nurses.

5. Complications

- **Atelectotrauma**
  This is caused by the intermitted collapse and re-expansion of the alveoli causing tissue damage e.g. with the use of too low PEEP.
- **Barotrauma**
  This is caused by the use of high pressure on the fragile lung; therefore the lowest pressure to achieve adequate ventilation should be used.
- **Oxygen toxicity**

The use of oxygen causes the creation of free radicals that damage the lungs and other tissues. Free radical damage is more likely with higher inspired oxygen but may also occur at relatively low oxygen concentrations, particularly with very immature infants who have reduced antioxidant defences. For preterm infants, increased FiO2 has also been associated with retinopathy of prematurity.

- **Pneumothorax**
- **Volutrauma**

This is caused by overdistension of the alveoli by the use of high lung volumes; therefore the lowest volume to achieve adequate ventilation should be used.

6. Humidity in Ventilated Neonates

Ideally ventilator gases should be as close to 37° centigrade and 100% humidity as possible at the point that they enter the baby. This corresponds to a water content of 44mg/l.

Consequences of too much or too little humidity

Potential consequences of dry or cool air	Potential consequences of excess humidity
• Increased heat loss • Increased insensible water loss • Drying and thickening of nasal and bronchial secretions • Cilial dysfunction resulting in poor clearance of secretions and increased risk of infection. • Inflammatory reaction of the lining cells of the respiratory tract. • Increased tendency to airway/ET tube plugging. • Bronchospasm • Possible increased risk of pneumothorax • Discomfort for the baby.	• If the ventilator gases are fully • humidified and warmer than body temperature, condensation could occur within the lungs. • Condensation occurring in the tubing can trickle back in to the airway or block the tubing causing false ventilator or CPAP pressure measurements. • Can cause auto-triggering of ventilation

Optimal Humidity

- The humidifier heats the gas to 37^{0}C and 100% relative humidity.
- The gas then passes into tubing that is heated by wire to 40^{0}C preventing any cooling or loss of humidity
- The last short piece of tubing beyond the temperature sensor is unheated, allowing the gas to cool back to 37^{0}C by the time it reaches the baby.
- Gas at 37^{0}C and 100% relative humidity is delivered to the baby
- The expiratory limb is unheated

- The gas cools rapidly on this side of the circuit and water condenses and should be collected in the trap before reaching the expiratory block of the ventilator.

The expiratory limb should pass down-hill from the patient to prevent water trickling back into the patient.

Problems with rain-out

- This is due to ambient air outside the tubing being cool, causing the water vapour inside the tubing to cool down and condense.
- Check the temperature probe is inserted firmly into the circuit.
- Check for circuits leaks/loose connections (sometimes the ventilator won't detect small leaks).
- Minimise draughts within the nursery.
- Avoid the use of fans near the ventilator circuit.
- Position tubing to minimise the risk of rain out running into the airway.

7. Indications for Intubation and Ventilation

If a baby either potentially or actually requires ventilation this is discussed consultant to consultant with Plymouth Neonatal Intensive Care Unit. A plan of treatment is made and retrieval is planned.

Indications for Intubation and Ventilation.	
In the delivery room.	In SCU
• Infants on CPAP but yet with worsening RDS symptoms.	• Stabilisation of infants at risk of sudden collapse, (e.g. apnoeas, severe sepsis, need to maintain airway patency).
• Failure to establish adequate respiration.	• Respiratory failure
Diaphragmatic hernia other surgical causes	• Pulmonary haemorrhage
Unstable Airway	• Elective situations e.g. transfer

8. Suggested initial settings

8.1. Various ventilator strategies are available to optimise gas exchange whilst minimising lung injury.

8.2. These are the default settings – PC-AC + VG for all babies ventilated on admission to the SCU. Different settings may be ordered by clinicians according to the gestational age and amount of support the baby needs.

8.3. Use PC-AC with VG Mode as default setting

Pmax The safe maximum pressure that the ventilator will not exceed to achieve calculated tidal volume. Usually set at 5 above the average PIP delivered to achieve the tidal volume. (Please set manually after 5 minutes)	25
Paw high (cms/H20)	30
PEEP (cms/H20) To prevent airway collapse	5.0
Inspiratory time (seconds) Inspiratory time and rate of breaths per minute should be such that full expiration is allowed.	0.34
Rate (bpm) The default is automatically set at 50. If appropriate this can be changed manually.	50
Tidal volume (mls/kg) calculated automatically when weight is entered	4-6mls/kg
Inspired Oxygen (FiO2)	Adjust according to clinical need
Slope (seconds)	0.08

9. Monitoring

9.1. All ventilated infants should be monitored as described by use of:-

- Pulse oximetry
- Capnography
- Flow sensor
- Blood Gas
- Chest x-ray - to confirm endotracheal tube position and lung inflation.
- ECG monitoring
- Blood pressure monitoring (non-invasive will suffice if arterial blood pressure monitoring is not in situ

- Temperature – regular axilla temperature checks, skin probe/ rectal probe use if appropriate

10. Initial target blood gas values

	Preterm	Term
pH	7.30 – 7.35	7.35 – 7.45
pC02	4.5 – 6.0 kPa	4.5 – 6.0 kPa
Arterial p02	5 – 8 kPa	7 – 10 kPa
Base excess	-4.0 to +4.0	-4.0 to +4.0

Exceptions to these values require discussion with a consultant.

Some clinical conditions will have differing blood gas expectations of values:

- Infants with severe chronic lung disease. (Permissive hypercapnia avoids over ventilation, and minimises injury through volutrauma and barotrauma)
- Infants with pulmonary hypertension

Frequency of Blood Gases

Blood gas analysis should be performed according to an infant's requirements.. Middle grade doctors and senior nursing staff will be able to advise on frequency of blood gas monitoring.

Alterations to ventilator settings or the administration of surfactant are a valid reason to check a gas 20-30 minutes later.

11. Considerations when changing ventilation settings

Look at the blood gas result

Consider:

- Does it fit with the baby's clinical picture
- Is there poor perfusion?
- Did the baby bleed easily?
- Was there a bubble in the specimen?
  Was it a capillary gas?

Consider taking an arterial sample

Look at the baby.

- Is the chest moving?
- What's the air-entry like?

- Is the baby struggling on the ventilator?
- Is the baby very tachypnoeic or is the baby apnoeic?

Look at the ventilator.

- Is it cycling?
- Are you giving the baby the ventilator settings you thought you were?
- What tidal volume (VT) is the baby getting?
- Is there a significant leak?
- Is it set up properly with an appropriate inspiratory time and with appropriate pressures?
- How stable has the baby been over the past few hours or days?
- Are there lots of secretions?
- How is the baby handling?

12. Changing Ventilation settings

When the settings are to be changed consider what is trying to be achieved.

Please note it is not possible to provide guidance for all eventualities, and if uncertain, the practitioner should always seek support from experienced senior personnel and staff working in Intensive Care.

See below for basic recommendations only.

Problem	Possible solution	Comment
Low oxygenation (Low PaO2 or saturations)	Increase the FiO2	• O2 increased by >10% then consider clinical exam and CxR • Check TV is being achieved. If not increase PMax • If not on VG Increase PIP – may also affect CO_2 • Increase inspiratory time – watch I:E ratio • Consider haemoglobin
High Oxygenation (High PaO2 or saturations)	Decrease the FiO2	• Decrease PEEP – if higher than 5cm • if not on VG decrease PIP – may affect CO_2 • Decrease inspiratory time
Over-ventilation (High pH with a low	Decrease the rate	If on VG:- • Decrease tidal volume (TV) • Decrease PMax

PaCO2)		If not on VG:- • Decrease the PIP • Decrease the rate – usually 5 -10 breaths (In Patient Trigger modes, this is futile if infant breathing above back up rate) • If on PC- AC change to SIMV if infant is breathing more than set rate (if no change then the rate can be reduced)
Under-ventilation (Low pH with a high PaCO2)	Increase the MAP Increase the rate	If on VG:- • If infant on VG, increase the TV If not on VG:- • Increase PIP to achieve good chest movement • Increase rate or inspiratory time – usually 5-10 breaths (Remember I:E ratio) • If on SIMV change to PC-AC if infant is breathing more than set rate or increase the rate

13. Deterioration

Consider DOPE

Any change in parameters should trigger a clinical assessment with the "DOPE"
Mnemonic as a prompt:

- **D**isplaced tube
- **O**bstructed tube
- **P**neumothorax
- **E**quipment failure

D - Displaced tube - Actions
- Auscultation of the chest.

- CO2 detector
- Check ventilator display - ? high leak or disconnected tube, or alarming as apnoea. Normal flow patterns, tidal volumes and minute volumes make displacement unlikely

O - Obstructed tube - Actions:

- Auscultation and suction via ET tube.
- Check ventilator which may also alarm with “tube obstruction” or “low MV” alarms and flow waves will be abnormally flat and minute volume will fall. Normal ventilator measures should suggest that obstruction is unlikely.

P – Pneumothorax - Actions:

- Assessment by “cold light” looking for transillumination of the chest on the side of the pneumothorax.
- Auscultation (not useful in very small preterm infants) cannot be used to exclude pneumothorax.
- Chest xray in more stable infants. This must not delay insertion of a chest drain in a deteriorating infant with “cold light” signs of pneumothorax.

E - Equipment failure - Actions:

- Neopuff the infant and assess for improvement.
- Check all the ventilation tubes are correctly inserted.

Note, for much of this assessment, disconnection from the ventilator is not necessary, and may lead to a reduction in information available, for example minute volume, tidal volume settings and flow settings. It also results in a loss of PEEP, which can cause a further deterioration.

14. Oxygen administration

The target oxygen saturations will vary depending on the reason for ventilating the infant. A premature infant will have lower target saturations than a term infant as they are at risk of oxygen toxicity and retinopathy of prematurity.

Term infants at risk of persistent pulmonary hypertension (e.g. those with meconium aspiration syndrome or hypoxic ischaemic encephalopathy), higher saturation targets are set (e.g. above 98%).

(See oxygen administration guidelines for neonates)

15. Over ventilation

Surfactant administration can dramatically change lung compliance and necessitate rapid changes in ventilator settings.

Whatever changes are made always remember to document the changes and inform the nurse looking after the infant.

16. Balanced ventilation

If one setting is very high consider balancing the settings e.g.

- If a baby is in 100% oxygen but with low pressures settings, it may be preferable to reduce the FiO2 but increase the pressures.
- If the baby is on high pressure settings but a low rate, it may be better to give a faster rate and lower pressures.

17. Hand ventilation

If the infant does require “hand ventilation” this should be using a pressure controlled system (e.g. Neopuff) to minimise barotrauma and to sustain PEEP wherever possible.

18. Management of the ventilated baby

- Consultation on management of the ventilated infant is directed by the tertiary centre, (Derriford NICU). All discussions are documented.
- Retrieval of the ventilated infant is arranged with the PNTS
- Hourly observations of infant’s clinical status and ventilator measurements (follow care plans). Report concerns immediately.
- Monitor blood gases as required and adjust ventilation requirements accordingly
- Monitor fluid balance hourly, check and work out 12 hourly (follow care plans)
- Medications (including analgesia and muscle relaxation)
- Minimal handling
- Positioning (see developmental care guidelines and care plan) placing the infant prone may improve the mechanics of breathing.
- Parental involvement - keep parents updated with information on condition and check their understanding. Discuss care plan and gain informed consent. Involve them in ‘cares’ where possible e.g. mouth care, containment holding.
- Ventilator circuits should be changed every 7 days or when visibly soiled or mechanically malfunctioning.

19. Extubation

Early extubation has been shown to result in a reduction in chronic lung disease.

Timing of extubation will depend on the clinical condition of the infant.

Babies may be extubated with a MAP of 7-8. (This should not be viewed as a target, as some infants may be extremely unstable with this MAP with others can be successfully extubated with a MAP as high as 12). Consider extubation in infants who clinically stable with low pressures as early as possible

To facilitate extubation infants:

- Discontinue morphine infusion
- Consider commencing caffeine (especially if under 30 weeks)
- Consider commencing CPAP or high flow oxygen
- Be prepared to re-intubate if clinically necessary

20. Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

20.1. Key performance indicators comprise:

- Nice Neonatal Quality Standards
- NHS Toolkit for High Quality Neonatal Services
- National Neonatal Audit Programme
- NHS Standard Contract for Neonatal Critical Care

Process for Implementation and Monitoring Compliance and Effectiveness

Staff are informed of revised documentation. There is an expectation that staff are responsible to keep updated on any improvements to practice and deliver care accordingly.

Activity for SCU including intensive care days are monitored using badger data base.

Data is also collected and monitored monthly via the South West Neonatal Network dashboard. Exception reporting occurs monthly via this dashboard for babies inappropriately cared for in SCU not transferred to a NICU. This data is compared across Neonatal Units.

Intensive care incidents are monitored by the neonatal governance team and neonatal network. Incidents are reported by the Datix system and South West Neonatal Network incident reporting process.

Undetected oesophageal intubation due to failure to monitor appropriately with capnography will be reported and investigated as a never event.

Infants cared for prolonged periods on the SCU will be exception reported to the Neonatal Network.

Non-adherence is reviewed and action plans made if required. Discussion and reviews occur at South West Neonatal Network governance meetings, Trust Directorate meetings, Governance meetings and Ward meetings. Learning and action plans are cascaded at these meetings and improvements implemented. Key findings and learning points will be disseminated across network and to relevant staff.

21. References

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22. Associated Documentation

- ABL 90 Blood gas analyser competency
- Chest Drain insertion, management and removal for pneumothorax in Neonates Guideline
- Developmental Care Guidelines for Neonates
- Elective Neonatal Intubation guidelines
- Continuous Positive Airway Pressure (CPAP) and Bi-level Continuous Positive Airway Pressure (BiPAP) Neonatal Nursing Care Guidelines
- Oxygen administration guidelines for neonates